

Jason Hong

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

B.S. in Computer Science and Engineering

Sept 2025 – May 2028

- GPA: 5.0/5.0
- Relevant Coursework: Computation Structures, Algorithms, Fundamentals of Programming
- Leadership: Associate Advisor and Orientation Leader

EXPERIENCE

Durability

Cambridge, MA

Software Engineer Intern

June 2026 – Aug 2026

- Built an end-to-end bulk roster-upload pipeline across React, Supabase/Deno, and SwiftUI, automating validation, batched invitations, email OTP, and first-login password setup.
- Prevented duplicate accounts during concurrent uploads with a single-flight database guard and retry boundaries, validated by 85 tests (64 UI, 21 integration).

Kyte Dynamics

Los Angeles, CA

Software Engineer Intern

May 2025 – Sept 2025

- Built and physically validated bidirectional custom MAVLink/uORB messaging between two PX4 flight controllers over UART, implementing outbound streaming, inbound decoding, and test publication.
- Refactored PX4 uORB interfaces for paired ODrive motor controllers and WitMotion IMUs, replacing duplicated left/right schemas with generic multi-instance topics; isolated defects across the uORB, protocol, and UART layers to bring both feedback instances online.
- Registered runtime PX4 parameters for commands, surfaced in QGroundControl to support driver-level ODrive testing.

RESEARCH

MIT Laboratory for Information & Decision Systems

Cambridge, MA

Undergraduate Researcher (UROP) | Advised by Prof. Tamara Broderick

April 2026 – Present

- Migrated [Nomon](#)'s language-model interface to the TextSlinger API to benchmark language models
- Diagnosed inconsistent conditioning between letter and word probabilities and corrected it in log space, reducing simulated correction rate from 3.99% to 0.68%.
- Designed output-validity probes that exposed 33.5% invalid word suggestions invisible to the oracle simulator; a vocabulary-fenced variant reduced junk to 0% and cut paired simulated click load from 1.390 to 1.147.

HONORS & AWARDS

MIT Pozen Fellowship: Awarded to select students for excellence in engineering and public service (2026)

Dr. Sherrie Emoto Mathematics Scholarship: Highest departmental honor for mathematical achievement at Fullerton College (2025)

PROJECTS

RISC-V Neural Network Inference Accelerator | *Minispec/BSV, C, RISC-V*

Spring 2026

- Optimized a pipelined processor and MNIST inference stack, achieving an **80×** improvement over baseline.
- Extended the ISA with custom instructions and pipelined SIMD multiplication; cut clock period from **1072 ps** to **529 ps** via a Kogge-Stone adder and by re-encoding ALU branch output as a single comparison bit, and added a specialized iCache to reduce CPI.

TECHNICAL SKILLS

Languages: C/C++, Python, Swift, Minispec, Bash

Tools & Frameworks: Linux/Unix, Git, Neovim, SwiftUI, React, Supabase, CMake, MAVLink

Interests: Computer Architecture, Verified Software Engineering, OS Kernels, Performance Engineering